

Recombine supplier-side search via innovation ambidexterity

An empirical study on Hong Kong manufacturing firms

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Abstract

Purpose – The purpose of this paper is to investigate the individual effects of boundary-spanning search from suppliers (supplier-side search (SS)). It is proposed that SS contributes to innovation ambidexterity (IA) and then business performance (BP). Further, this paper includes buyer–supplier relationships (BSRs) and competitive intensity (CI) as moderators to clarify boundary conditions.

Design/methodology/approach – An ordinary least squares regression was employed to test hypotheses, based on 184 sets of data from Hong Kong manufacturing firms. The SPSS version of PROCESS was utilized.

Findings – The results show that IA partially mediates the relationship between SS and BP. Contingently, the direct effect is negatively moderated by BSRs and CI.

Research limitations/implications – This paper confirms the partial mediating effect of IA on the relationship between SS and BP. Additional mediators, such as organizational innovation and marketing innovation, can be examined in the future.

Practical implications – This paper contributes to practice by suggesting that suppliers are a rewarding single source for firms to undertake boundary-spanning search. IA plays a significant role in reinforcing the effects of SS on BP and should be implemented with sustaining efforts. BSRs and CI can be detrimental and should be responded to cautiously.

Originality/value – This paper highlights the individual effects of SS on BP. Furthermore, the underlying process and boundary conditions are analyzed. The exploitation-exploration framework is applied throughout the entire study.

Keywords Competitive intensity, Buyer–supplier relationships, Innovation ambidexterity, Empirical survey, Exploitation-exploration framework, Supplier-side search

Paper type Research paper



Introduction

The commercialization of heterogeneous knowledge contributes critically to improved performance and continued survival in firms (Zhou *et al.*, 2014). Firms do not always possess the necessary knowledge within their boundaries (Filieri and Alguezaui, 2014). This is especially the case in the context of highly competitive transition economies (Handfield and Bechtel, 2002). To survive in this knowledge era, firms are typically urged to shift toward open innovation by establishing knowledge flow channels with external sources, such as suppliers, customers and universities (Hipple, 1988; Chesbrough, 2003). However, this practice can oftentimes be burdensome for firms to optimally cover several sources (Katila and Ahuja, 2002; Laursen and Salter, 2006). Alternatively, a lower number of sources, and in many cases only one, will be sought to counterbalance costs (Laursen and Salter, 2006; Inkpen and Tsang, 2016). Manufacturers often resort to upstream suppliers for knowledge acquisition (Choi and Krause, 2006; Wagner and Bode, 2014). For example, Geely, a Chinese automobile manufacturer, designs new car models by introducing technology from its suppliers (Wang *et al.*, 2016b). Moreover, Un *et al.* (2010) empirically document that suppliers, rather than customers or universities, are the most promising knowledge source for manufacturing firms.

Channel structures can be described as being arm's length, contractual or collaborative (Heide, 1994), which play important roles when it comes to suppliers voluntarily transferring knowledge to a buyer firm (Wagner and Bode, 2014). Specifically, knowledge acquisition can be achieved by a firm through the knowledge sharing that is initially pushed by its suppliers when they are in collaboration. Alternatively, knowledge acquisition can also be achieved through the supplier-side search (SS) with the buyer firm being the more active party (Yam *et al.*, 2011). Knowledge sharing with suppliers has been well recognized as a best practice for buyer product development (Lawson *et al.*, 2009), and is significantly beneficial for buyer firm performance (Lau, Tang and Yam, 2010; Lau *et al.*, 2010b). However, SS is relatively underemphasized. Although boundary-spanning search has been highly advocated (Rosenkopf and Nerkar, 2001; Cousins *et al.*, 2011), managers still struggle to identify which individual channel is rewarding when it is too costly for firms to cover many (Laursen and Salter, 2006). With this in mind, this study seeks to empirically verify whether suppliers are a profitable single source from which a firm conducts boundary-spanning search.

Knowledge components that are acquired by boundary-spanning search will not always guarantee performance improvement in a firm (Cousins *et al.*, 2011; Ferreras-Méndez *et al.*, 2015). Katila and Ahuja (2002) document that the extent to which a firm search widely for new knowledge is positively related to firm performance. On the contrary, Sidhu *et al.* (2007) find no significant relationship between supply-side search (specifically, technology search) and firm performance and Laursen and Salter (2006) confirm that conducting deep search from an excessive number of sources leads to lower firm performance. Inspired by these mixed results, this study further reviews and summarizes that: first, redundant knowledge, even in a large number, has a limited effect on firm performance (Caner *et al.*, 2017). Thus, the relationship between SS and business performance (BP) may be attributed to both the number and the variety of knowledge components (Yayavaram and Ahuja, 2008; Caner *et al.*, 2017). Second, the resources offered by suppliers cannot be effectively leveraged without right conditions (Jajja *et al.*, 2017) and the resources, if not rare and inimitable enough, bring few values (Barney, 1991). Thus, the impact of SS on BP may work through some mediating process, such as innovation ambidexterity (IA) (Zhang *et al.*, 2016). Third, the motivation for and efficiency of knowledge transfer are influenced by several contingent factors (Gao *et al.*, 2015). Thus, the relationship between SS and BP may be contingent on specific

contextual factors, such as buyer-supplier relationships (BSRs) and competitive intensity (CI). Accordingly, this study develops three research questions:

RQ1. How does SS influence BP?

RQ2. What is the role of IA in the relationship between SS and BP?

RQ3. How do BSRs and CI affect the relationship between SS and BP?

To address these research questions, this study uses the exploitation-exploration framework that involves quantity, refinement and efficiency, while involving variation, creation and effectiveness on the other hand (March, 1991; Rothaermel and Alexandre, 2009). Fittingly, SS coincides with the exploitation-exploration framework along with different amounts of learning in nonlocal domains and thus with acquired knowledge being composed of the supplementary and the complementary (Knudsen, 2007; Sidhu *et al.*, 2007). Furthermore, the underlying mechanisms behind SS and BP are also in line with the exploitation-exploration framework, from the perspectives of components and architectures. For one thing, heterogeneous components are acquired, which can depend on both supplementary knowledge for increasing number and complementary knowledge for increasing variety (Rodan and Galunic, 2004; Sammarra and Biggiero, 2008). Thus, SS contributes to BP directly (Yayavaram and Ahuja, 2008; Caner *et al.*, 2017). For another, idiosyncratic architectures are generated by IA that work to relink knowledge components, which can rely first on recombinant reuse for relinking supplementary knowledge with refined architectures and second on recombinant creation for relinking complementary knowledge with created architectures (Carnabuci and Operti, 2013). Thus, SS contributes to BP indirectly via IA (Lin *et al.*, 2013; Zhang *et al.*, 2016). Moreover, BSRs and CI as moderators influence the number and the variety of obtained components (Gao *et al.*, 2015).

The empirical results of this study verify that SS contributes to BP directly and indirectly via IA. Further, the direct effect is negatively moderated by BSRs and CI. The contributions are summarized as follows. First, the exploitation-exploration framework is utilized to analyze the SS as well as the underlying mechanisms behind its direct and indirect effects on BP, which extends the application of this framework (Rothaermel and Alexandre, 2009). Second, SS is found to be beneficial for BP. This finding responds to studies that call for the investigation of individual channels for boundary-spanning search (Laursen and Salter, 2006; Ferreras-Méndez *et al.*, 2015). Third, IA is introduced as the intermediate link between SS and BP, which enriches the research on innovation value chain (Roper *et al.*, 2008; Ganotakis and Love, 2012) and the mediating role of IA (Lin *et al.*, 2013; Zhang *et al.*, 2016). Finally, this study enriches the research on BSRs and CI by examining their moderating roles (Ang, 2008; Villena *et al.*, 2011).

The remainder of this paper is organized as follows. The next section elaborates theoretical background, followed by hypothesis development. Methodological section outlines sample and data collection, questionnaire design and measures, and measurement validation. The section that follows presents hypothesis testing and results. Finally, this paper concludes with theoretical and managerial implications, as well as the limitations and future research.

Theoretical background

Open innovation and SS

Open innovation can be divided into two groups that are inbound innovation and outbound innovation (Busarovs, 2013) and this study is included in the former. Firms shift to open innovation for knowledge acquisition by knowledge sharing and boundary-spanning search (Yam *et al.*, 2011). Knowledge sharing with suppliers usually takes place in the form of supplier integration that is conducted to help a receiving firm develop innovative products (Corsten and Felde, 2005; Luzzini *et al.*, 2015). Lawson *et al.* (2009) document that knowledge

sharing with suppliers is driven by socialization and ends up increasing buyer firm performance. Lau, Tang and Yam (2010) and Lau *et al.* (2010b) argue that supplier integration contributes to product performance, which is mediated by product innovation and modularity. Contingently, Corsten and Felde (2005) find that the relationship between supplier integration and buyer performance is moderated by trust and dependence.

Boundary-spanning search is typically characterized by depth and breadth (Laursen and Salter, 2006). Lopez-Vega *et al.* (2016) further extend the search space by introducing heuristics. Yam *et al.* (2011) document that scanning for external information works to benefit innovation capability and in turn sales performance. Contingently, Sidhu *et al.* (2007) find that the effects of supply-side search, demand-side search and geographic search on firm performance are contingent on environmental dynamisms. However, few studies focus on the particularities of boundary-spanning search within each individual channel, such as suppliers (Ferreras-Méndez *et al.*, 2015). In this study, SS refers to the boundary-spanning search that is conducted by a firm from its suppliers individually (Laursen and Salter, 2006). As knowledge sharing with suppliers has been highly advocated, SS is also worth such attention and further investigation. Table I structurally demonstrates this research gap.

Exploitation-exploration framework

The exploitation-exploration framework is widely applied in many fields, such as knowledge search and technological innovation (Rothaermel and Alexandre, 2009). Sidhu *et al.* (2007) posit that nonlocal search is the indicative of exploitation and exploration with different amounts of learning. This observation should be true of the SS. Acquired knowledge can be classified into the supplementary and the complementary, with the former being characterized by a high level of redundancy and the latter being less redundant (Knudsen, 2007). Accordingly, by conceptualizing search space as the relative distance from a firm's knowledge base (Levinthal and March, 1993; Lavie *et al.*, 2010), the supplementary knowledge is acquired when SS denotes adjacent exploitation, while the complementary knowledge is acquired when SS denotes distant exploration (Knudsen, 2007; Lopez-Vega *et al.*, 2016). Analyzing the effects of SS in terms of these two kinds of knowledge will present more details.

Additionally, the mechanisms underlying the effects of SS on BP also follow the exploitation-exploration framework. From the perspective of components, firms search intensively for knowledge components to form effective combinations for performance

Articles	Types of knowledge acquisition		Sources of knowledge		Different sources are analyzed individually or integrally	
	Search	Sharing	Suppliers	Multiple sources	Individually	Integrally
This paper	✓		✓		✓	
Un <i>et al.</i> (2010)		✓		✓	✓	
Corsten and Felde (2005)		✓	✓		✓	
Lawson <i>et al.</i> (2009)		✓	✓		✓	
Lau, Tang and Yam (2010)		✓	✓		✓	
Lau <i>et al.</i> (2010b)		✓	✓		✓	
Luzzini <i>et al.</i> (2015)		✓	✓		✓	
Laursen and Salter (2006)	✓			✓		✓
Sidhu <i>et al.</i> (2007)	✓			✓		✓
Cousins <i>et al.</i> (2011)	✓			✓		✓
Yam <i>et al.</i> (2011)	✓			✓		✓
Ferreras-Méndez <i>et al.</i> (2015)	✓			✓		✓
Lopez-Vega <i>et al.</i> (2016)	✓			✓		✓

Table I.
Empirical research on
knowledge acquisition

problem solving (Caner *et al.*, 2017). Drawing on the NK model, the number of knowledge components (N) and the distinctions among them (K) determine the effectiveness of knowledge combinations (Yayavaram and Ahuja, 2008; Caner *et al.*, 2017). Accordingly, supplementary knowledge and complementary knowledge can contribute to more effective combinations by increasing the number and the variety of knowledge components.

From the perspective of architectures, the ways in which different components are linked together determine the effectiveness of combinations (Henderson and Clark, 1990). To relink supplementary knowledge and complementary knowledge, firms work on reusing routine and creating new recombinant patterns (Carnabuci and Operti, 2013). Exploitative innovation (EI) builds on the refinement of existing technological trajectories for existing markets, while exploratory innovation (ER) involves departure from existing technological trajectories to cater for new markets (Benner and Tushman, 2003; Jansen *et al.*, 2006). Further, EI and ER can be engaged in simultaneously and maintained at a high level, i.e., IA (Tushman, 1996; Lin *et al.*, 2013; O'reilly and Tushman, 2013). In this vein, supplementary knowledge and complementary knowledge trigger recombinant reuse and recombinant creation, which contribute to IA. Thereafter, refined and novel architectures are generated for more effective combinations (Henderson and Clark, 1990; Barney, 1991; Zhang *et al.*, 2016).

Buyer–supplier relationships and competitive intensity

Social network theory emphasizes that network content, network structure and surrounding condition are all necessary for a comprehensive understanding (Gao *et al.*, 2015). For this study, the supply chain can be regarded as a simplified network. SS reflects the relational content, for which the value relies on the number and the variety of knowledge components. BSRs and CI represent the relational structure and surrounding condition, respectively, which have influences on the number and the variety (Gao *et al.*, 2015).

BSRs refer to the social closeness between a firm and its suppliers, characterized by varying degrees of trustworthiness and commitment (Zhou *et al.*, 2014; Thomas *et al.*, 2015). Past studies have confirmed the beneficial aspects of BSRs by reducing interfirm distance (Dyer and Singh, 1998). Recent developments pay more attention to the detrimental respects of BSRs by considering rigidity and opportunism (Gilbert, 2005; Villena *et al.*, 2011). CI refers to the severity of competition in a particular industry (Porter, 1985). Heavy competition generates unpredictable changes and puts firms in vulnerable positions (Ang, 2008; Zhou *et al.*, 2014). Evidences have shown that organizational outcomes are negatively affected by competitive forces (Barnett, 1997; Chowdhury and Sarkar, 2017). Accordingly, this study sheds light on the negative moderating roles of BSRs and CI. For clarity, Figure 1 outlines the conceptual model of this study.

Hypothesis development

The impact of SS on BP

It is postulated that SS is positively related to BP. Consistent with the logic of NK model, the positive effects of SS on BP can be attributed to the number (N) and the variety (K) of knowledge components (Yayavaram and Ahuja, 2008; Caner *et al.*, 2017). In terms of supplementary knowledge, the overlapping knowledge increases relatedness between acquired and internal

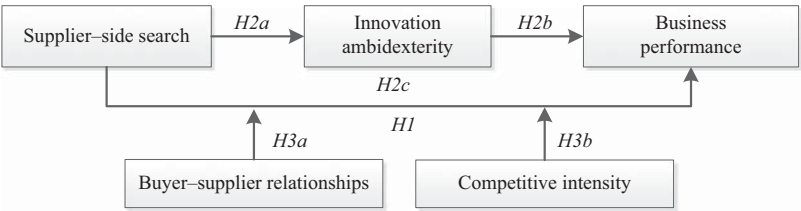


Figure 1.
Conceptual model

knowledge (Kim *et al.*, 2015). This favors knowledge acquisition for three reasons. First, firms can clearly distinguish between the relevant and the irrelevant regarding what is familiar to them (Riedl *et al.*, 2013). Upon realizing this, they will pragmatically judge whether the components are worthwhile of further efforts (Wang *et al.*, 2016a). Second, firms that possess related components with suppliers will develop more relation-specific assets to make knowledge transfer more efficient (Kotabe *et al.*, 2003). Third, more related routines between a firm and its suppliers streamline the process of knowledge internalization (Wang *et al.*, 2016a). Therefore, supplementary knowledge acquired by SS can easily reach a larger magnitude (N).

In terms of complementary knowledge, the uniqueness facilitates “shake-up” in a firm’s existing routines and thereby establishes novel pathways (Caner *et al.*, 2017). For one thing, exposure to novel perspectives and dissimilar knowledge components breaks a firm’s path dependence to some degrees, which make the firm more inclusive in distant search (Levinthal and March, 1993; Laursen, 2012). For another, the successful reception of complementary knowledge leads to the creation of new projects and utilization of the knowledge for distinctive purposes (Knudsen, 2007), which further increases the firm’s inclusiveness. With this in mind, it is not impossible for a firm to acquire complementary knowledge by SS and thereby increase the variety (K). With numerous supplementary knowledge and various complementary knowledge, SS contributes to increased BP:

H1. SS has a positive impact on BP.

The mediating role of IA

Innovation requires not only access to but also the recombination of knowledge components (Sammarra and Biggiero, 2008). For this study, it is postulated that SS triggers multiple types of recombinant patterns that facilitate IA. In terms of supplementary knowledge, it triggers the reuse of routine recombinant patterns, thereby contributing to EI. This is first because the ability for a firm to learn from its specific partner increases with the number of overlapping knowledge (Dyer and Singh, 1998; Kim *et al.*, 2015). Such a firm can understand the supplementary to a deeper extent and can easily find feasible recombinant patterns within existing technological trajectories. Second, in line with path dependence elaborated by Laursen (2012), recombinant patterns are typically considered among those familiar to a firm, when it comes to supplementary knowledge. Thus, it is more likely for a firm to recombine supplementary knowledge by reusing routine recombinant patterns (Carnabuci and Operti, 2013).

In terms of complementary knowledge, it triggers the creation of new recombinant patterns, thereby contributing to ER. For one thing, even though firms typically prefer recombinant reuse, the recombinant patterns necessary to relink complementary knowledge do not always coincide with a firm’s familiarity (Laursen, 2012). That is to say, it is not so achievable for a firm to stick to its internal routines to recombine complementary knowledge, as would be the case for supplementary knowledge. If so, the firm will be supposed to create new recombinant patterns (Carnabuci and Operti, 2013). For another, complementary knowledge triggers higher learning potential, offering a platform for creating new recombinant patterns (Knudsen, 2007). Under this condition, firms will have greater motivation to engage in recombinant creation. Taken together, SS that triggers both recombinant reuse and recombinant creation provides a firm with opportunities to refine and depart from existing technological trajectories. Such opportunities work to foster EI and ER simultaneously, that is, to generate IA (Lin and McDonough, 2011):

H2a. SS is positively associated with IA.

In turn, IA is assumed to be favorable for BP. Savino *et al.* (2017) take innovation as the fundamental factor for BP. Specifically, EI helps firms to strengthen their footholds in existing markets through higher efficiencies, making them more profitable and empowering

them to offer lower prices (Jansen *et al.*, 2006). At the same time, ER helps firms to nurture new sources of competitive advantage and establish presences in new and emerging markets through creative activities, making them more differentiated (Lavie *et al.*, 2010). Therefore, IA allows firms with low-cost and differentiation advantages to adapt to competitions in the long run (Cao *et al.*, 2009; Voss and Voss, 2013).

In promoting IA, EI and ER are supportive of each other, which contribute to superior performance (Cao *et al.*, 2009; Lavie *et al.*, 2010). First, EI facilitates a firm's profitability with higher reliability, which provides ER with reliable financial backing (Lavie *et al.*, 2010). Second, EI enhances a firm's capability within existing technological trajectories, which lays the foundation for capability development in complementary fields of technology (Cohen and Levinthal, 1990; Ferreras-Méndez *et al.*, 2015). In reverse, successful ER empowers firms with new technologies, which facilitates EI to increase efficiencies in complementary domains (Cao *et al.*, 2009; Voss and Voss, 2013). ER also enriches a firm's internal knowledge base, making EI take effect on a larger scale (Knudsen, 2007; Cao *et al.*, 2009). In light of these considerations, a positive relationship is expected between IA and BP:

H2b. IA is positively related to BP.

Finally, IA is assumed to be a mediator between SS and BP. For one thing, the resources offered by suppliers cannot be effectively leveraged without right conditions (Jajja *et al.*, 2017). The innovation value chain studies show that the knowledge that is sourced by a firm will be translated into innovation, thereby generating superior performance (Roper *et al.*, 2008). Similarly, this study takes IA as the intermediary link between SS and BP.

For another, organizational resources lead to superior performance only if they are recombined to be rare and inimitable enough (Barney, 1991; Lin *et al.*, 2013; Zhang *et al.*, 2016). Following this logic, when acquired components are relinked with idiosyncratic architectures that are shaped by IA, knowledge combinations will be more particular for the focal firm, thus contributing to superior performance (Cousins *et al.*, 2011; Ferreras-Méndez *et al.*, 2015). Specifically, EI refines existing knowledge architectures to relink supplementary knowledge into conventional combinations, which are efficient for solving performance problems in adjacent fields of technology and then empower the firm with a low-cost advantage (Katila and Ahuja, 2002; Jansen *et al.*, 2006). ER creates novel knowledge architectures to relink complementary knowledge into distinctive combinations, which are effective for solving performance issues in distant fields of technology and then contribute to an advantage in differentiation (Katila and Ahuja, 2002; Lavie *et al.*, 2010):

H2c. IA mediates the relationship between SS and BP.

The contingent role of BSRs

It is posited that BSRs weaken the direct effect of SS on BP due to reduced number (N) and less variety (K) in knowledge components. First, with increasing closeness of BSRs, a buyer firm gradually takes suppliers' daily activities and operations for granted, and thus fails to detect suppliers' changes (Villena *et al.*, 2011). This eases the formation of unchanged resource investment, i.e., resource rigidity (Gilbert, 2005) and then makes the buyer firm less motivated to increase search efforts. As a result, the number of acquired components decreases. Simply put, N decreases.

Second, the relational closeness to existing suppliers increases the buyer firm's path dependence and its satisfaction with existing yet potentially inferior solutions (Villena *et al.*, 2011). This eases the formation of unchanged organizational process, i.e., routine rigidity (Gilbert, 2005) and then makes the buyer firm rely more on repeated search activities, rather than creative search efforts (Levinthal and March, 1993). As a result, the obtained components tend to be more redundant. Simply put, K decreases.

Furthermore, close BSRs may bring about potential supplier opportunism, due to the frequently occurring trust asymmetry (Graebner, 2009; Villena *et al.*, 2017). Where a buyer firm trusts its suppliers, the suppliers may not reciprocate in their moral obligations. In this vein, the buyer firm may increase its relational investments or reduce its monitoring over time (Skowronski and Benton, 2018). This offers the window of opportunity through which suppliers can conceal the valuable knowledge or reveal repeated knowledge (Liu *et al.*, 2009), leading to decreases in both the number and the variety of obtained components:

H3a. The relationship between SS and BP is negatively moderated by BSRs.

The contingent role of CI

It is postulated that CI weakens the direct effect of SS on BP due to reduced number (*N*) and less variety (*K*) in knowledge components. As competition intensifies, firms tend to focus more on their own survival (Jansen *et al.*, 2006). In this case, suppliers may be inclined to protect their internal knowledge of value from spillover by offering less timely or less accurate feedback and forgoing other unnecessary obligations, which slows interfirm knowledge transfer (Zhou *et al.*, 2014). Therefore, the number of acquired knowledge components decreases. Simply put, *N* decreases.

On the other hand, with intensified competition, a firm and its suppliers tend to focus more on efficiency-oriented activities (Jansen *et al.*, 2006). They shift their priorities to have more reliability and predictability (Katila and Ahuja, 2002). As a result, the knowledge bases of suppliers are less diverse and the search patterns of the buyer firm are more repetitive. This makes the knowledge components that are acquired by SS more redundant. Simply put, *K* decreases:

H3b. The relationship between SS and BP is negatively moderated by CI.

Research methodology

Sample and data collection

This study's data were collected from Hong Kong manufacturing firms located at Dongguan, a prefecture-level city in Guangdong province of mainland China. The list of sample firms was obtained from the Dongguan Association of Enterprises with Foreign Investment (DAEFI). These sample firms are members of the DAEFI and thus have obligations to complete questionnaires organized by the association. This survey was supported by the Dongguan municipal government and thus the questionnaire was distributed as the attachment to an official letter, further encouraging sample firms to complete and return the questionnaires. Follow-up telephone calls and mailings were used to improve the response rate. To reduce social desirability bias, the survey claimed that all individual responses would be kept anonymous, with exhibited results limited to aggregate statistics.

The initial sample consists of 330 firms and data collection efforts yield submissions from 289 firms, with response rate of 87.6 percent. Among these returned questionnaires, 95 questionnaires were excluded due to over many blank items. Additionally, ten questionnaires were excluded, as they were not from manufacturing firms. The final data set includes 184 questionnaires, with the effect rate of 56 percent. Most of the respondents were at least junior managers that had been working in their current positions for more than three years. Their average job tenure was 9.3 years. Thus, it is reasonable to assume that these respondents were knowledgeable about their firms' operations and innovative activities. The profile of the 184 sample firms and the respondents was shown in Table II.

As there was one respondent per firm, the common method bias was assessed. Following the work of Flynn *et al.* (2010), Harman's single-factor test was first conducted (Podsakoff *et al.*, 2003). The results reveal that there are six factors with eigenvalues greater than or equal to 1.0, which explains 69.3 percent of the total variance. The first factor explains 28.7 percent

Table II.
Profile of the sample

Firm profile	Count	%	Cumulative %
<i>Number of employees</i>			
$X < 20$	3	1.63	1.63
$20 \leq X < 300$	107	58.15	59.78
$300 \leq X < 1,000$	57	30.98	90.76
$X \geq 1,000$	17	9.24	100.00
<i>Annual sales (in million RMB)</i>			
$Y < 3$	35	19.02	19.02
$3 \leq Y < 20$	62	33.70	52.72
$20 \leq Y < 400$	83	45.11	97.83
$Y \geq 400$	4	2.17	100.00
<i>Industry type</i>			
Textile and apparel	38	20.65	20.65
Furniture manufacturing	2	1.09	21.74
Paper making and paper products	7	3.80	25.54
Printing and recording medium reproduction	1	0.54	26.09
Culture and education, arts and crafts, sports and recreational equipment manufacturing	4	2.17	28.26
Chemical	3	1.63	29.89
Rubber and plastic products	15	8.15	38.04
Ferrous metal smelting and rolling processing	1	0.54	38.59
Metal products	26	14.13	52.72
Ordinary machinery	7	3.80	56.52
Equipment for special purposes	8	4.35	60.87
Electrical machinery and equipment	26	14.13	75.00
Electronic equipment manufacturing	8	4.35	79.35
Instrumentation manufacturing	3	1.63	80.98
others	35	19.02	100.00
<i>Respondent profile</i>			
Chief executive officer	14	7.61	7.61
General manager	35	19.02	26.63
Assistant general manager	17	9.24	35.87
Middle manager	78	42.39	78.26
Junior manager	18	9.78	88.04
Others	22	11.96	100.00

Note: $n = 184$

lower than the 50 percent cut-off. Second, a confirmatory factor analysis was applied to Harman's single-factor model. The model's fit indices, specifically $\chi^2(377) = 2,420.043$, $\chi^2/\text{df} = 6.419$, CFI = 0.425, TLI = 0.381, IFI = 0.429 and RMSEA = 0.172, are unacceptable and far worse than those for the measurement model. This indicates that a single factor is not acceptable. Third, a measurement model that includes only traits, as well as a method factor model that includes both the traits and one method factor, was tested (Podsakoff *et al.*, 2003). The results of the method factor model improve fitness by 0.001 (CFI = 0.938, TLI = 0.930, IFI = 0.939), with the common method factor accounting for 18.49 percent of the total variance. The path coefficients for the trait factors and their significances are similar between these two models. This suggests that the measurement model is robust, despite the inclusion of one method factor. All results of these tests provide an evident indication that there is no need for concerns regarding common method bias.

Late response bias was tested by the differences between early respondents and late respondents regarding variables of the hypotheses. Average values found in the survey's

responses for the first 10 percent of the respondents were compared with those of the last 10 percent. The results of the *t*-test show no statistical significance, indicating that late response bias is not a serious concern (Lau *et al.*, 2010b).

Questionnaire design and measures

An extensive survey of literature was conducted to identify appropriate definitions and measures for constructs. Because the constructs and items are adapted from the literature in English while respondents are Chinese, the questionnaire was first developed in English and then translated into Chinese. Thereafter, the Chinese version was translated back into English by two independent researchers and then a third researcher worked with the aforementioned two researchers to check for discrepancies between the back-translated English version and the original one. These were done to ensure conceptual equivalence.

Two business leaders from the DEAFI, who were familiar with both the survey's objectives and the general situation of these sample firms, were invited to assess content validity. Some items were reworded, specifically from academic language into industry terms, in order to be accurately interpreted by the respondents. To refine the measurement further, a pilot test was conducted with 30 Hong Kong manufacturing firms. Accordingly, minor modifications on item description were made.

Independent and dependent variables

Supplier-side search. According to the work of Laursen and Salter (2006), SS refers to the extent to which a firm draws intensively from its suppliers. In this study, it is measured using the observed variable communication frequency in the process of new product development as a proxy. More frequent communication can lead to more effective communication (Reagans and McEvily, 2003), which is pretty important in knowledge transfer (Ferrerias-Méndez *et al.*, 2015; Caner *et al.*, 2017). Therefore, the communication frequency works well to indicate the extent to which a firm draws knowledge from its suppliers.

Business performance. This variable is measured by five items adapted from the work of Li and Atuahene-Gima (2001). Respondents were asked to assess their firms' BP using a seven-point Likert scale (1 = much worse, 7 = much better). The perceptual measurement of BP is utilized in this study for multiple reasons. First, objective performance data are unavailable from financial databases because these firms are self-sustaining entities with independent accounting systems. Second, sample firms are reluctant to share objective data due to concerns regarding confidentiality (Corsten and Felde, 2005). Third, as for companies differing in size and market segments, resources are different in each case and thus it is not appropriate to employ objective measurements to rate firm performance (Lau *et al.*, 2010a).

Mediating variable

Innovation ambidexterity. This variable is measured in two steps. First, EI and ER are measured separately by six respective items adapted from the work of Jansen *et al.* (2006). Second, the two measures are combined together using an additive approach to measure IA (Lubatkin *et al.*, 2006; Zhang *et al.*, 2016). Respondents were asked to assess the EI and ER using a five-point Likert scale (1 = strongly disagree, 5 = strongly agree).

Moderating variables

Buyer-supplier relationships. This variable is measured by four items adapted from the work of Anderson and Weitz (1992) and Siguaw *et al.* (1998). Respondents were asked to assess the BSRs using a five-point Likert scale (1 = strongly disagree, 5 = strongly agree).

Competitive intensity. This variable is measured by six items adapted from the work of Jaworski and Kohli (1993). Respondents were asked to assess the CI using a five-point Likert scale (1 = strongly disagree, 5 = strongly agree).

Covariates

This study involves four covariates: firm size, firm age, R&D intensity and technological turbulence (TT). Firm size (the number of employees) and firm age are both included as controls after logarithm transformation. R&D intensity is measured by the proportion of innovation investments to annual sales. TT is measured by six items adapted from the work of Jaworski and Kohli (1993). Respondents were asked to assess the TT using a five-point Likert scale (1 = strongly disagree, 5 = strongly agree).

Measurement validation

Confirmatory factor analysis (CFA) was conducted to assess the reliability and validity of the constructs. The goodness of fit indices for measurement model ($\chi^2/df = 1.625$, CFI = 0.937, TLI = 0.929, ILI = 0.938, RMSEA = 0.058) is acceptable. The results of the construct measure and CFA are reported in Table III, including measurement items, factor loading, standard error (SE), Cronbach's α , composite reliability (CR) and average variance extracted (AVE). The instruments for constructs are validated by exploratory factor analysis (specifically, a principle component analysis with varimax orthogonal rotation). The results presented in Table IV reveal 29 measurement items and six factors, further confirming the structure of the constructs (Lau, Tang and Yam, 2010).

Reliability. The values for Cronbach's α , ranging from 0.715 to 0.938, are above the 0.70 cut-off. Additionally, the values for CR, ranging from 0.741 to 0.939, are also larger than the 0.70 cut-off (Fornell and Larcker, 1981b). These results demonstrate a satisfactory reliability.

Convergent validity. All items are loaded significantly on their corresponding latent constructs, with acceptable values for standardized factor loading ranging from 0.48 to 0.92. All but one of these values are greater than the 0.50 cut-off (Chin, 1998). Moreover, the values for AVE, ranging from 0.501 to 0.755, are all above the 0.50 cut-off (Fornell and Larcker, 1981a). These results confirm an acceptable convergent validity.

Discriminant validity. The same-factor loadings on the constructs they measure are larger than the cross-factor loadings on the other constructs they are not meant to measure (Chin, 1998). The square roots of the AVE are greater than all corresponding correlations, and all correlations are smaller than their respective values for reliability (Fornell and Larcker, 1981a). These results provide evidences for an acceptable discriminant validity.

Analysis and results

Table V presents mean values and standard deviations of all variables as well as their correlation matrix. Prior to hypothesis analysis, the variance inflation factors for all regression models were calculated and the results are far lower than 10. This indicates that there is no need for concerns regarding multicollinearity. In this study, the simple effects (*H1*, *H2a* and *H2b*) are tested using the hierarchical regression analysis. The mediating and moderating effects (*H2c*, *H3a* and *H3b*) are tested based on the SPSS version of PROCESS, an ordinary least squares regression path analysis modeling tool (Hayes, 2013).

Table VI presents the results of hierarchical regression analysis. Therein, Equation (2) shows that SS has a positive and significant impact on BP ($\beta = 0.200$, $p < 0.01$), supporting *H1*. Equation (6) shows that SS is positively and significantly related to IA ($\beta = 0.248$, $p < 0.05$), supporting *H2a*. Equation (3) shows that the relationship between IA and BP is positive and significant ($\beta = 0.205$, $p < 0.01$), supporting *H2b*.

Measurement items	Factor loading	SE
<i>Business performance</i> ($\alpha = 0.938$; CR = 0.939; AVE = 0.755)		
BP1: return on investment	0.92	–
BP2: return on sales	0.88	0.050
BP3: return on assets	0.89	0.048
BP4: Sales growth	0.85	0.054
BP5: market share growth	0.80	0.056
<i>Exploitative innovation</i> ($\alpha = 0.911$; CR = 0.909; AVE = 0.626)		
EI1: we frequently refine the provision of existing products	0.78	0.075
EI2: we regularly implement small adaptations to existing products	0.81	0.073
EI3: we introduce improved, but existing products for our local market	0.74	0.091
EI4: we improve our provision's efficiency of products.	0.85	–
EI5: we increase economies of scales in existing markets	0.82	0.074
EI6: our unit expands products for existing clients	0.74	0.072
<i>Exploratory innovation</i> ($\alpha = 0.932$; CR = 0.934; AVE = 0.704)		
ER1: our unit accepts demands that go beyond existing products	0.74	0.074
ER2: we invent new products	0.88	0.055
ER3: we experiment with new products in our local market	0.90	–
ER4: we commercialize products that are completely new to our unit	0.89	0.055
ER5: we frequently utilize new opportunities in new markets	0.83	0.059
ER6: our unit regularly uses new distribution channels	0.78	0.068
<i>Buyer-supplier relationships</i> ($\alpha = 0.715$; CR = 0.741; AVE = 0.501)		
BSRs1: we defend this supplier when outsiders criticize the company	0.73	0.139
BSRs2: we are not continually lookout for another supplier to replace or to add to our current supplier	0.86	–
BSRs3: we are patient with suppliers when they make mistakes that cause us trouble (<i>D</i>)	–	–
BSRs4: we are willing to dedicate whatever people and resources it takes to grow sales for this supplier	0.48	0.111
<i>Competitive intensity</i> ($\alpha = 0.842$; CR = 0.841; AVE = 0.515)		
CI1: competition in our industry is cutthroat	0.67	0.094
CI2: there are many "promotion wars" in our industry (<i>D</i>)	–	–
CI3: anything that one competitor can offer, others can match readily	0.73	0.092
CI4: price competition is a hallmark in our industry	0.67	0.104
CI5: one hears of a new competitive move almost every day	0.81	–
CI6: our competitors are relatively strong	0.70	0.086
<i>Technological turbulence</i> ($\alpha = 0.847$; CR = 0.838; AVE = 0.571)		
TT1: the technology in our industry is changing rapidly	0.65	0.078
TT2: technological changes provide big opportunities in our industry	0.82	0.071
TT3: it is very difficult to forecast where the technology in our industry will be	0.60	0.072
TT4: a large number of new product ideas have been made possible through technological breakthroughs in our industry	0.91	–

Table III.
Results of
the construct
measurement
and confirmatory
factors analysis

Notes: $n = 184$. (*D*) denotes that the item was eliminated during the process of construct validation

Test of mediation

The method elaborated by Baron and Kenny (1986) can be used to informally judge if the proposed mediation occurs (Zhang *et al.*, 2016). For this study, Equation (4) in Table VI shows that IA has a positive and significant effect on BP ($\beta = 0.182$, $p < 0.01$), with SS being controlled. This result, together with the test results of *H1*, *H2a* and *H2b*, suggests that the mediating effect of IA on the relationship between SS and BP is likely to occur.

Table IV.
Results of exploratory
factor analysis

Items	Rotated component matrix Components					
	BP	EI	ER	BSRs	CI	TT
BP3	0.917	0.098	0.086	0.013	0.054	0.006
BP1	0.905	0.138	0.147	0.038	-0.015	0.026
BP2	0.872	0.184	0.065	0.007	-0.011	0.136
BP4	0.866	0.090	0.155	0.118	0.026	0.085
BP5	0.849	0.124	0.096	0.052	-0.011	0.068
EI1	0.056	0.828	0.200	0.181	0.065	0.007
EI4	0.146	0.801	0.280	0.007	0.133	-0.003
EI2	0.134	0.782	0.286	0.115	0.144	0.063
EI5	0.193	0.729	0.418	0.029	-0.004	-0.041
EI6	0.103	0.706	0.304	-0.054	-0.009	0.016
EI3	0.245	0.635	0.323	0.089	0.092	0.152
ER2	0.138	0.239	0.831	0.100	0.100	0.052
ER4	0.117	0.236	0.824	0.075	0.090	0.213
ER3	0.156	0.280	0.800	0.141	0.156	0.081
ER5	0.121	0.324	0.768	0.061	0.126	0.110
ER6	0.095	0.245	0.747	0.054	0.159	0.251
ER1	0.061	0.335	0.680	-0.006	0.051	0.194
BSRs2	0.009	0.170	0.045	0.831	0.089	0.077
BSRs1	0.140	0.176	0.002	0.779	0.042	0.043
BSRs4	0.071	-0.202	0.298	0.553	0.179	0.182
CI5	0.034	0.050	0.104	0.084	0.792	0.256
CI3	-0.024	0.076	0.073	0.026	0.787	0.072
CI4	-0.038	-0.002	0.042	-0.012	0.781	-0.005
CI6	0.072	0.030	0.084	0.034	0.717	0.174
CI1	-0.029	0.033	-0.009	0.215	0.680	0.174
TT1	0.084	0.038	0.105	0.088	0.131	0.770
TT2	0.084	-0.056	0.319	0.030	0.109	0.769
TT4	0.083	0.019	0.262	0.020	0.280	0.768
TT3	0.049	0.149	0.028	0.068	0.180	0.763

Notes: $n = 184$. Extraction method: principal component analysis; Rotation method: varimax with Kaiser normalization

More formally, a bootstrapping technique was employed to test the mediating effect based on the PROCESS (Preacher and Hayes, 2004; Zhang *et al.*, 2016). Consulting Hayes's templates, this study utilized the model 4 to test the mediation (Hayes, 2013). The bootstrap subsamples were set as 5,000 and the confidence level for confidence interval was set at 95% ($CI_{95\%}$) (Zhao *et al.*, 2010). The test results are shown in Table VII. Therein, the indirect effect on BP from SS via IA is positive and significant ($\beta = 0.044$, $SE = 0.021$, $CI_{95\%} = [0.012, 0.099]$), supporting $H2c$. Furthermore, the direct effect of SS on BP is still positive and significant ($\beta = 0.155$, $SE = 0.063$, $CI_{95\%} = [0.030, 0.280]$). Thus, IA partially mediates the relationship between SS and BP, namely complementary mediation (Zhao *et al.*, 2010).

Test of moderation

The model 5 was utilized to test the moderating effects of BSRs ($H3a$) and CI ($H3b$) on the relationship between SS and BP. Taking marginal significance into account, the confidence level for confidence interval was set at 90% ($CI_{90\%}$) (McClelland and Judd, 1993). Table VIII presents the test results of the moderating effect of BSRs. The second submodel, which includes the interaction of SS and BSRs, shows that the coefficient from this interaction to BP is negative and significant ($\beta = -0.135$; $CI_{90\%} = [-0.257, -0.013]$). Thus, $H3a$ is supported. Moreover, different values for BSRs lead to different claims about the conditional

Table V.

Descriptive statistics
and correlation matrix

Variables	BP	SS	EI	ER	BSRs	CI	Size	Age	R&D	TT
BP	0.869									
SS	0.244***	na								
EI	0.336***	0.202***	0.791							
ER	0.296***	0.256***	0.627***	0.839						
BSRs	0.086	0.061	0.105	0.190***	0.707					
CI	0.018	0.062	0.135*	0.222***	0.199***	0.718				
Size	0.151**	0.108	0.121	0.123*	0.067	-0.011	na			
Age	-0.056	-0.074	-0.025	-0.092	-0.004	-0.146**	0.103	na		
R&D	-0.008	0.225***	0.072	0.245***	-0.048	0.045	0.067	0.049	na	
TT	0.190***	0.088	0.214***	0.412***	0.117	0.357***	0.036	-0.123*	0.079	0.756
Means	3.409	3.200	3.777	3.466	3.422	3.651	2.316	2.517	2.180	3.223
SD	1.089	1.250	0.857	1.033	0.906	0.883	0.488	0.637	1.226	0.949

Notes: $n = 184$. The square of the construct's AVE is provided along the diagonal. Significance levels are two tailed, with * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$

	Business performance			Innovation ambidexterity		
	Equation (1)	Equation (2)	Equation (3)	Equation (4)	Equation (5)	Equation (6)
Constant	2.262***	1.855***	1.603**	1.363**	3.207***	2.703***
Age	-0.095	-0.061	-0.072	-0.048	-0.111	-0.068
Size	0.325**	0.276*	0.246	0.217	0.386	0.324
R&D	-0.022	-0.067	-0.067	-0.097	0.221**	0.165*
TT	0.227**	0.215**	0.116	0.119	0.539***	0.524***
BSRs	0.079	0.063	0.032	0.025	0.228*	0.209
CI	-0.087	-0.091	-0.111	-0.111	0.115	0.110
SS		0.200***		0.155**		0.248**
IA			0.205***	0.182***		
R^2	0.067	0.116	0.151	0.180	0.186	0.217
Adjusted R^2	0.036	0.081	0.118	0.142	0.159	0.186
F-value	2.125*	11.844***	19.591***	25.410***	6.762***	13.618**

Table VI.

Regression results of
the simple effects

Notes: $n = 184$. Significance levels are two tailed with * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

direct effects of SS on BP (Hayes, 2013). As shown in Table VIII, the conditional direct effects are significant at the low level ($\beta = 0.282$, $CI_{90\%} = [0.127, 0.437]$) and at the moderate level ($\beta = 0.160$, $CI_{90\%} = [0.056, 0.264]$), but not significant at the high level of BSRs ($\beta = 0.038$, $CI_{90\%} = [-0.111, 0.186]$). Going into more details, Johnson–Neyman technique offers moderating values (i.e. JN) to define significance regions (Hayes, 2013). In this study, the JN(BSRs) equals to 3.783. Specifically, when BSRs are less than or equal to 3.783, the conditional direct effects of SS on BP are significant. Otherwise, SS cannot significantly predict BP. In brief, the direct effect of SS on BP vanishes with BSRs. This provides further supports to *H3a*.

Similarly, Table IX presents the test results of the moderating effect of CI. The second submodel, which includes the interaction of SS and CI, indicates that the coefficient from this interaction to BP is negative and significant ($\beta = -0.125$; $CI_{90\%} = [-0.230, -0.019]$). Thus, *H3b* is supported. Furthermore, the conditional direct effects are significant at the low level ($\beta = 0.252$, $CI_{90\%} = [0.120, 0.384]$) and at the moderate level ($\beta = 0.142$, $CI_{90\%} = [0.037, 0.246]$), but not significant at the high level of CI ($\beta = 0.032$, $CI_{90\%} = [-0.115, 0.179]$). The JN(CI) equals to 3.903. Specifically, when CI is less than or equal to 3.903, the conditional direct effects of SS on BP are significant. Otherwise, SS cannot significantly predict BP. In brief, the direct effect of SS on BP vanishes with CI. This provides further supports to *H3b*.

Table VII.
Regression results of
the mediation analysis

	Innovation ambidexterity		Business performance	
Model 4	Coeff. (SE)	[LL95%CI, L95%CI]	Coeff. (SE)	[LL95%CI, UL95%CI]
Constant	2.703 (0.974)	[0.780, 4.626]	1.363 (0.652)	[0.077, 2.650]
IA			0.182 (0.049)	[0.084, 0.279]
SS	0.248 (0.095)	[0.061, 0.435]	0.155 (0.063)	[0.030, 0.280]
Age	−0.068 (0.183)	[−0.430, 0.293]	−0.048 (0.120)	[−0.285, 0.189]
Size	0.324 (0.237)	[−0.143, 0.792]	0.217 (0.156)	[−0.091, 0.524]
R&D	0.165 (0.096)	[−0.025, 0.355]	0.097 (0.063)	[−0.223, 0.028]
TT	0.524 (0.130)	[0.269, 0.780]	0.119 (0.089)	[−0.056, 0.294]
BSRs	0.209 (0.129)	[−0.046, 0.464]	0.025 (0.085)	[−0.143, 0.193]
CI	0.110 (0.141)	[−0.168, 0.389]	−0.111 (0.093)	[−0.294, 0.072]
	$R^2 = 0.217$		$R^2 = 0.180$	
	$F(7,176) = 6.967, p < 0.01$		$F(8,175) = 4.789, p < 0.01$	
	Total, direct, and indirect effects between supplier-side search and business performance			
	Coeff. (SE)		[LL95%CI, UL95%CI]	
Total effect	0.200 (0.064)		[0.073, 0.327]	
Direct effect	0.155 (0.063)		[0.030, 0.280]	
Indirect effect	0.044 (0.021)		[0.012, 0.099]	
Notes: $n = 184$. LL, lower limit; UL, upper limit. Unstandardized regressions coefficients are reported				

Table VIII.
Regression results of
the moderating
effects of BSRs

	Innovation ambidexterity		Business performance	
Model 5	Coeff. (SE)	[LL90%CI, UL90%CI]	Coeff. (SE)	[LL90%CI, UL90%CI]
Constant	3.978 (0.918)	[2.460, 5.496]	1.987 (0.637)	[0.934, 3.040]
IA			0.173 (0.049)	[0.091, 0.254]
SS	0.257 (0.095)	[0.100, 0.414]	0.160 (0.063)	[0.056, 0.264]
BSRs			0.021 (0.085)	[−0.119, 0.161]
SS × BSRs			−0.135 (0.074)	[−0.257, −0.013]
Age	−0.059 (0.184)	[−0.363, 0.245]	−0.051 (0.119)	[−0.248, 0.147]
Size	0.349 (0.237)	[−0.044, 0.742]	0.206 (0.155)	[−0.050, 0.463]
R&D	0.153 (0.096)	[−0.006, 0.312]	−0.105 (0.063)	[−0.210, −0.001]
TT	0.535 (0.130)	[0.320, 0.750]	0.146 (0.089)	[−0.001, 0.294]
CI	0.150 (0.140)	[−0.081, 0.381]	−0.113 (0.092)	[−0.265, 0.039]
	$R^2 = 0.205$		$R^2 = 0.195$	
	$F(6,177) = 7.623, p < 0.01$		$F(9,174) = 4.687, p < 0.01$	
	Coeff. (SE)		[LL90%CI, UL90%CI]	
BSRs	Conditional direct effects of supplier-side search on business performance			
Mean−SD (2.516)	0.282 (0.094)		[0.127, 0.437]	
Mean (3.422)	0.160 (0.063)		[0.056, 0.264]	
Mean + SD (4.328)	0.038 (0.090)		[−0.111, 0.186]	
IA	Indirect effect of supplier-side search on business performance			
	0.044 (0.021)		[0.018, 0.088]	
Notes: $n = 184$. Both supplier-side search and BSRs are mean centered before product				

The interaction plots in terms of BSRs and CI are separately presented in Figure 2. It is shown that the higher the values for BSRs and CI, the lower the slopes. That is to say, SS has a weaker direct effect on BP, when BSRs are closer and when CI is higher, consistent with *H3a* and *H3b*.

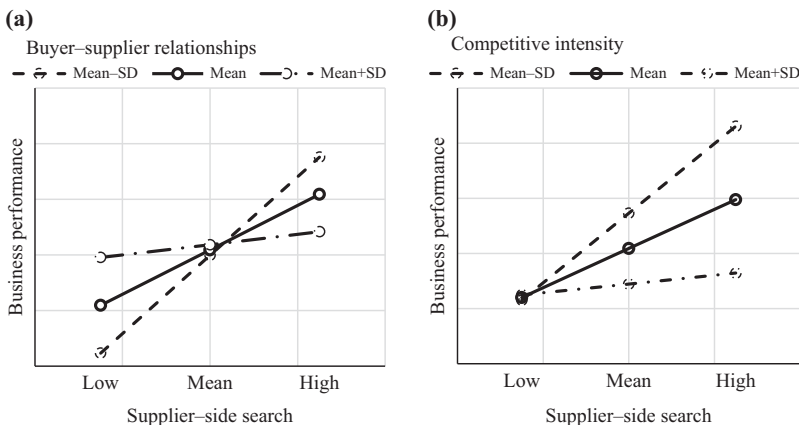
Discussion and conclusions

Theoretical implications of this study are fourfold. First, the exploitation-exploration framework is embedded throughout the entire study. Specifically, SS is indicative of both

	Innovation ambidexterity		Business performance	
Model 5	Coeff. (SE)	[LL90%CI, UL90%CI]	Coeff. (SE)	[LL90%CI, UL90%CI]
Constant	3.778 (0.890)	[2.307, 5.250]	1.467 (0.618)	[0.446, 2.488]
IA			0.178 (0.049)	[0.097, 0.259]
SS	0.249 (0.095)	[0.092, 0.405]	0.142 (0.063)	[0.037, 0.246]
CI			−0.145 (0.094)	[−0.300, 0.009]
SS × CI			−0.125 (0.064)	[−0.230, −0.019]
Age	−0.084 (0.182)	[−0.385, 0.216]	−0.086 (0.121)	[−0.285, 0.113]
Size	0.319 (0.237)	[−0.072, 0.710]	0.242 (0.155)	[−0.015, 0.498]
R&D	0.168 (0.096)	[0.009, 0.326]	−0.104 (0.063)	[−0.208, −0.000]
TT	0.557 (0.122)	[0.355, 0.760]	0.131 (0.088)	[−0.015, 0.277]
BSRs	0.227 (0.127)	[0.016, 0.437]	0.035 (0.085)	[−0.105, 0.175]
	$R^2 = 0.214$		$R^2 = 0.197$	
	$F(6,177) = 8.044, p < 0.01$		$F(9,174) = 4.750, p < 0.01$	
	Coeff. (SE)		[LL90%CI, UL90%CI]	
Competitive intensity	Conditional direct effects of supplier-side search on business performance			
Mean−SD (2.768)	0.252 (0.080)		[0.120, 0.384]	
Mean (3.651)	0.142 (0.063)		[0.037, 0.246]	
Mean + SD (4.534)	0.032 (0.089)		[−0.115, 0.179]	
IA	Indirect effect of supplier-side search on business performance			
	0.044 (0.021)		[0.016, 0.087]	

Notes: $n = 184$. Both supplier-side search and competitive intensity are mean centered before product

Table IX.
Regression results of
the moderating
effect of
competitive intensity



Notes: (a) The conditional direct effects by BSRs; (b) the conditional direct effects by competitive intensity

Figure 2.
Plots of the
conditional direct
effects of supplier-side
search on business
performance

exploitation and exploration (Sidhu *et al.*, 2007), thus with supplementary knowledge and complementary knowledge being acquired (Knudsen, 2007). Furthermore, the exploitation-exploration framework emerges as an appropriate theoretical perspective for explaining the underlying mechanisms behind the direct and indirect effects of SS on BP, thus with knowledge heterogeneity and recombination being analyzed (Carnabuci and Operti, 2013; Caner *et al.*, 2017). Therefore, this study extends the application of the exploitation-exploration framework.

Second, this study offers specific insights on SS and confirms its direct and positive effect on BP (*H1*). Although boundary-spanning search has been acknowledged as being

effective in obtaining external knowledge (Rosenkopf and Nerkar, 2001; Katila and Ahuja, 2002; Cousins *et al.*, 2011), little academic attention has been paid to the particularities within each individual channel (Ferrerias-Méndez *et al.*, 2015). This study confirms that suppliers are such an individual channel and are indeed a rewarding single source for firms to conduct boundary-spanning search.

Third, this study confirms the mediating effect of IA on the relationship between SS and BP (*H2a–H2c*). The innovation value chain studies emphasize the role of innovation in transforming knowledge search into performance improvement but narrow innovation types into simply product and process innovation (Roper *et al.*, 2008; Ganotakis and Love, 2012). This study contributes to the research on innovation value chain by introducing IA as an intermediary, something that has long been required in complex and dynamic environment (Zhang *et al.*, 2016). Additionally, extending studies that emphasize the impact of IA on firm performance (Cao *et al.*, 2009; Voss and Voss, 2013), this study provides an evidence on how IA acts as a facilitator for firms to benefit from SS (Lubatkin *et al.*, 2006; Lin *et al.*, 2013).

Fourth, this study finds that the relationship between SS and BP is negatively moderated by BSRs and CI (*H3a* and *H3b*). For one thing, this study enriches the research on BSRs. Past studies initially focus on the beneficial aspects of BSRs (Handfield and Bechtel, 2002). However, due to increasing awareness of trust asymmetry and supplier opportunism, increasing studies have turned to the detrimental aspects of BSRs (Mitrega and Zolkiewski, 2012; Obal, 2015). This is also true of the direct effect of SS on BP, which is negatively moderated by BSRs. For another, this study also enriches the research on CI. Although past studies assert that competition can motivate firms to work closely with their suppliers (Jansen *et al.*, 2006; Gao *et al.*, 2015), this study stands out in finding that CI negatively moderates the direct effect of SS on BP, which is consistent with the argument that fierce competition is always detrimental for firms' profitability (Barnett, 1997; Ang, 2008; Chowdhury and Sarkar, 2017).

Managerial implications

First, when covering multiple external sources is unaffordable for a firm, SS can be a feasible choice. Specifically, firms should make efforts to acquire both supplementary and complementary knowledge that are critical to enrich their knowledge base. In this vein, firms should particularly pay attention to the heterogeneity of their suppliers' knowledge. By engaging in boundary-spanning search from suppliers whose knowledge components are more heterogeneous, the buyer firm's knowledge base will be enriched to a greater degree.

Second, not only should firms pay attention to SS, but also they should engage in IA to generate idiosyncratic architectures. Specifically, firms should reuse routine recombinant patterns to relink their supplementary knowledge with refined architectures, while creating new recombinant patterns to relink their complementary knowledge with novel architectures. In this way, conventional combinations and distinctive combinations are both formed to solve multiple types of performance problems as a result.

Third, firms should understand how to match SS with BSRs and CI. For one thing, firms should be prudent and vigilant to foster close relationships with suppliers, and make certain efforts to attract suppliers' relationship-specific investments to maintain a balanced level of supplier-dependence on the buyer firm. For another, firms should take efforts to increase suppliers' motivation for knowledge transfer and exploration when confronted with highly fierce competition, or otherwise to switch to other sources that are not directly influenced by this competition.

Limitations and future research

This study is subject to some limitations that offer avenues for future research. First, IA partially mediates the relationship between SS and BP. Future research can further

explore other mediators, such as organizational innovation and marketing innovation, to formulate a more comprehensive understanding. Second, the manufacturing context of this study may limit the generalizability of the findings. According to the work of Laursen and Salter (2006), universities are the most important single source for biotechnology industry and lead users play the key role in scientific instruments industry. Future research can collect data from other industries to validate these findings. Finally, this study was inherent in cross-sectional research design, which prohibits causal inference. Future research can take efforts to collect longitudinal data to test the model in this study.

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